No single technology excels at all three since the laws of physics impose fundamental tradeoffs. High data rates require more power and work over shorter distances whereas long range communication at low power means accepting lower data rates.

3.3.2 Short-range technologies (<100m)

When sensors are relatively close to your data collection point and you need frequent updates or rich data, short-range technologies excel. The three dominant short-range technologies shown in table 3.3 each evolved to solve different problems. WiFi emerged from the need for high-speed computer networking and is fast and ubiquitous, but power-hungry. Bluetooth Low Energy (BLE) is the opposite, optimizing for battery life at the expense of range and throughput. Zigbee and Thread (together with the Matter interoperability standard) occupy a middle ground, designed specifically for sensor networks that need to scale beyond simple point-to-point connections by adopting a mesh-based network topology.

Table 3.3 Popular short-range wireless communication technologies compared on data rate, range, and power draw.

Technology	Data rate	Typical range	Power draw	Infrastructure needs	Home use case
WiFi	54Mbps-1Gbps+	30-100m	High (constant power)	Existing router	Indoor environmental sensors near outlets
Zigbee/Thread	250kbps	10-100m	Low	Dedicated hub	Considered but not selected
BLE	2Mbps	10-30m	Very low	Smartphone/hub	Electricity meter pulse counter

3.3.3 Long-range technologies (>100m)

When sensors are scattered across large areas, lack access to wired power supplies, or there is significant interference (for example, structural steel in an